CO-1

ASSASSMENT TOOL-2

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QUESTIONS :-

1. wins a spectrum auction for 45 MHz
   Company policy sets aside 25 kHz

   45 MHz = 45000 KHz
   45000 % 25 =1800
   1800 % 2 (paired to duplex)

   So the telecom operator gets 900 full-duplex channels.

2. Total number of voice channels = 560
   Cell frequency shared = 7
   560 % 7 = 80
   80 channels per cell

   So each cell gets 80 voice channels.

3. GSM operator's simultaneous voice channels
   100 carriers × 8 voice time slots = 800 simultaneous voice channels
   So the deployment can offer 800 simultaneous voice channels.

4. Serving signal compared with co-channel interference
   serving signal at -60 dBm
   co-channel interference at -92 dBm
   C/I = −60 − (−92) = 32 dB
   So the drive test shows a carrier-to-interference ratio of 32 dB.

6. operator holds 42 MHz
assigning 300 kHz per channel
42 MHz = 42,000 kHz
42,000 ÷ 300 = 140 total channels
140 ÷ 7 = 20 channels per cell
So each cell gets 20 channels.

7. With path-loss exponent 4, doubling the distance divides received power by $2^4 = 16$.
64 W ÷ 16 = 4 W
So the receiver should get about 4 W.

11. TDMA frame lasting 5 ms
equal time slots 8
TDMA system's slot duration
5 ms ÷ 8 = 0.625 ms = 625 μs
So each time slot lasts 0.625 ms.

12. Spectrum planners' carriers
24 MHz = 24,000 kHz
24,000 ÷ 200 = 120 carriers
So the allocation supports 120 carrier frequencies.

13. Power received by the base station
8 W = 39.03 dBm
39.03 − 33 = 6.03 dBm ≈ 6 dBm
So the base station receives approximately 6 dBm.
15. Cluster's call-handling capacity
6 cells × 250 calls per cell = 1,500 calls
So the cluster can handle up to 1,500 simultaneous calls.

16. Switching subsystem's active calls

3 minutes = 180 seconds

Active calls = 4,000 calls/second × 180 seconds = 720,000 calls

So about 720,000 calls could be active at once.

18. Spectrum wasted on guard bands

There are 29 gaps between 30 adjacent bands.

29 × 40 kHz = 1,160 kHz = 1.16 MHz

So 1.16 MHz is used for guard bands.

1 9. Vehicle's handoffs during an hour

Cell crossing distance = diameter = 2 × 1.5 = 3 km

At 90 km/h, crossing time = 3 ÷ 90 hours = 1/30 hour = 2 minutes

60 ÷ 2 = 30 handoffs

So the vehicle would have roughly 30 handoffs in an hour.

20. Network's total user capacity

8 regional centers × 40 base stations × 250 users = 80,000 users

So the network can serve up to 80,000 users.

22. FDMA system's channels

6 MHz = 6,000 kHz

6,000 ÷ 30 = 200 channels

So the allocation can provide 200 channels.

23. Market's offered traffic

Call duration = 2.5 ÷ 60 hours
Traffic = 1,200 × 4 × (2.5 ÷ 60) = 200 Erlangs

So the subscribers generate 200 Erlangs of offered traffic.

24. Deployment's simultaneous users

40 MHz = 40,000 kHz
Each user needs 400 kHz
40,000 ÷ 400 = 100 users

So the band can serve up to 100 users at once.

25. Carrier's total user capacity

1,200 base stations × 600 users = 720,000 users

So the network can support up to 720,000 simultaneous users.

26. Paging scheme's worst-case rounds

200 cells ÷ 4 cells per round = 50 rounds

So it takes 50 paging rounds in the worst case.

28. Subscribers roaming onto partner networks

600,000 × 15% = 90,000 subscribers

So about 90,000 subscribers are roaming at a given moment.

30. Operator's channels per cell

48 MHz = 48,000 kHz
48,000 ÷ 200 = 240 total channels
240 ÷ 4 = 60 channels per cell

So each cell gets 60 channels.

31. GSM deployment's channels

30 MHz = 30,000 kHz
30,000 ÷ 200 = 150 channels

So the spectrum supports 150 channels.

32. Successful handoffs

2,500 × 99% = 2,475 successful handoffs

So 2,475 calls were successfully handed off.

33. mobile handset radiates 12 W

link experiences 45 dB

12 W = 40.79 dBm
40.79 − 45 = −4.21 dBm ≈ −4.2 dBm

So the base station detects approximately −4.2 dBm.

34. Cluster's user capacity

7 cells × 900 users per cell = 6,300 users

So the cluster can serve up to 6,300 simultaneous users.